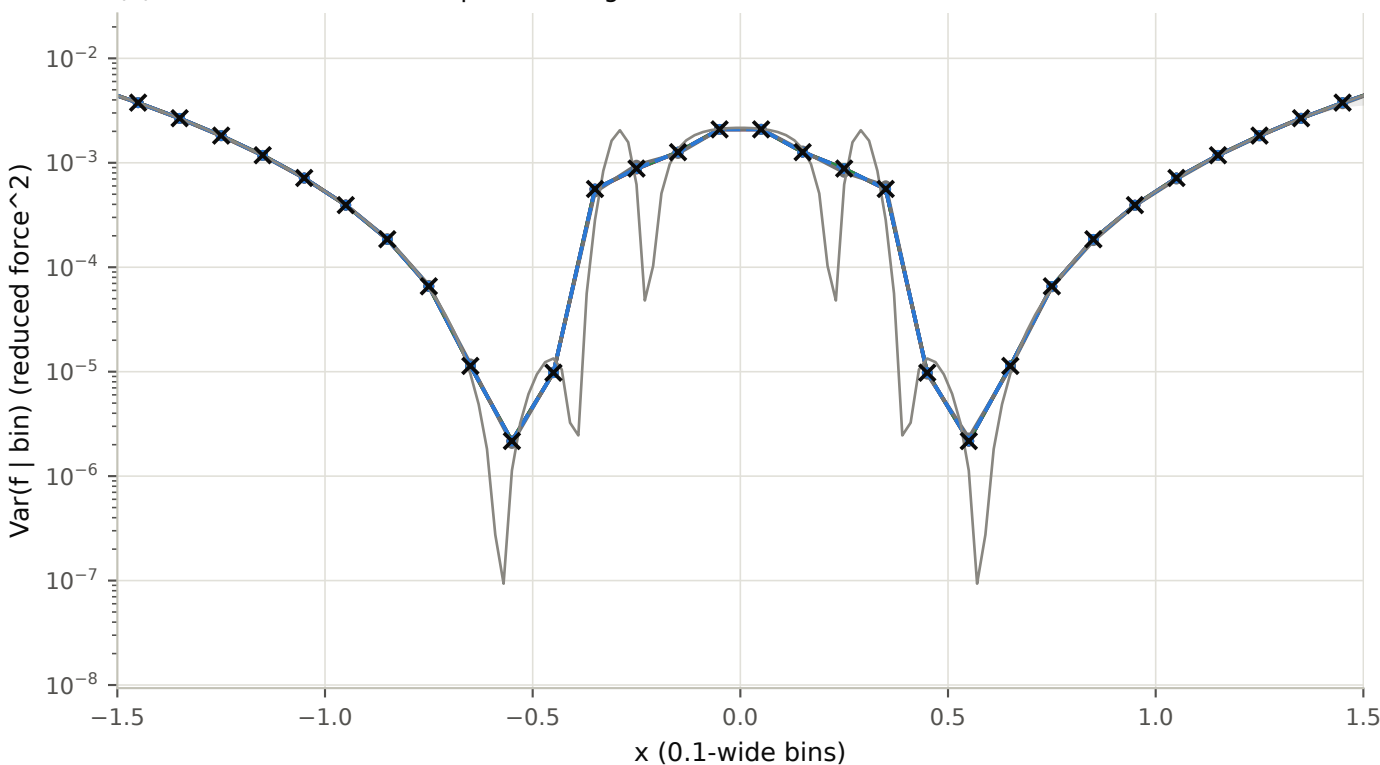
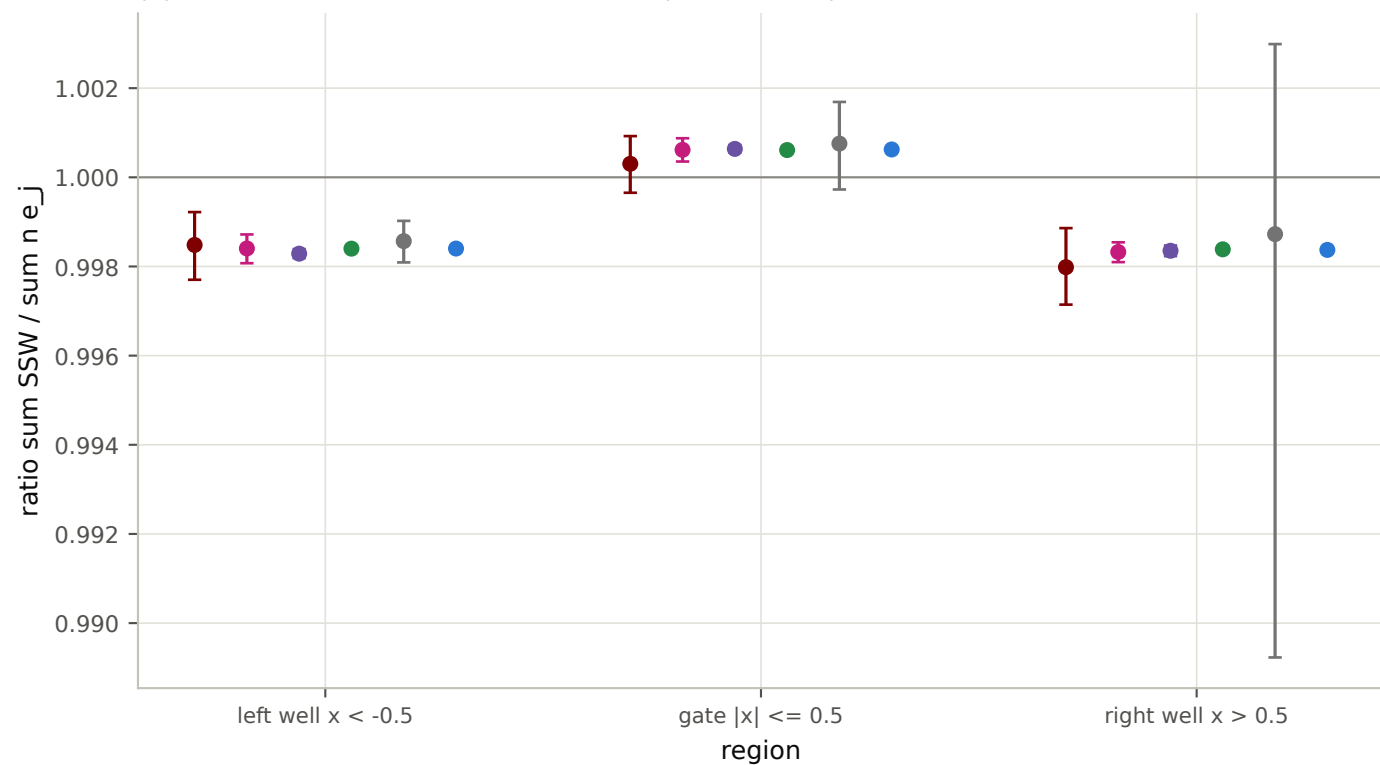


D2(iii) class-wise conditional force variance:  $l: \alpha = 0$  (purely energetic barrier in  $x$ ; deterministic force),  $N = 128$   
 Within-bin variance of the deposited force pooled over seeds vs the exact  $(2\alpha^2/\beta^2)(w'/w)^2$  (shift:  $(2/\beta^2)(w'/w)^2$ ) bin average plus the within-bin variance of  $F^*$ ; bands (a) and error bars (b): 95 % seed-bootstrap CIs.  
 EXPLORATORY: this quantity is not preregistered (SCIENTIFIC\_PLAN sections 4 and 6)

(a) within-bin variance of  $f$  per clone-age class



(b) measured / exact within-bin variance (eval window)



— clone age  $[0,0.01]$  t.u.    — clone age  $[0.1,1]$  t.u.    — never cloned    x exact  $e_j$ , deposit-weighted over the 0.1 bin (compare with the curves)    ..... exact  $\langle \text{Var}(f|x) \rangle_{\text{bin}}$   
— clone age  $[0.01,0.1]$  t.u.    — clone age  $\geq 1$  t.u.    — ABF (all deposits)    — exact  $e_j = \langle \text{Var}(f|x) \rangle + \text{Var}(F^* | \text{bin}), 0.02$  bins     (a, b) 95 % seed-bootstrap CI